Test Plan

Project Information

Project Name: End-to-End Testing for Sauce Demo Web & Mobile Application.
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Number of User Stories: 5 user stories.
Estimated Effort (Hours): 38 hours .

Project Description: This project focuses on performing end-to-end software testing on the Sauce Demo platform, including both web and mobile applications. The main goal is to ensure the system's quality and reliability by validating its core functionalities. The testing covers key areas such as user login, product browsing, cart management, and checkout process. Different testing types are applied, including functional, regression, and usability testing. The project also includes creating test scenarios and test cases, executing tests, and reporting defects to ensure a smooth and reliable user experience.

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Introduction

1.1 Purpose

The purpose of this test plan is to define the testing approach and ensure the quality and functionality of the Sauce Demo application.

1.2 System Overview

Sauce Demo is an e-commerce web and application that allows users to log in, browse products, add items to the cart, and complete the checkout process.

2. Scope

2.1 In Scope

- Login functionality .
- Product listing .
- Add to cart .
- Remove from cart .
- Checkout process.

2.2 Out of Scope

- Payment gateway functionality (external integrations) .
- Database testing (data validation at backend level) .
- Security testing (advanced security checks) .
- Admin or backend functionalities (if not accessible).

3. Test Objectives

The objective of testing is to verify that all core features are working correctly, detect defects, and ensure a smooth user experience.

4. Test Strategy

The testing strategy defines the overall approach used to validate the Sauce Demo application and ensure its quality and functionality.

4.1 Test Levels

Testing will be performed at the following levels:

- **System Testing:** Verifying the complete system functionality as a whole.
- **User Acceptance Testing (UAT):** Ensuring the system meets user requirements and expectations.

4.2 Test Types

In this project, the following testing types will be applied:

- **Manual Testing** will be used to execute all test cases manually.
- **White Box Testing** will be used to validate internal code logic and structure.
- **Black Box Testing** will be used to validate system functionality based on requirements.
    - **Functional Testing** to ensure all features work correctly.
    - **Non-Functional Testing:** Will evaluate system quality (performance, usability, security, compatibility) and UI/UX to ensure a smooth, user-friendly experience.

- **Automated Testing** will be used to execute repetitive and regression test cases using automation tools, ensuring faster execution and higher accuracy in test results.

4.3 Test Approach

The testing process will include both manual and automated testing:

- Manual testing will be used for exploratory testing and validating new features.
- Automation testing will be used for repetitive and regression test cases.
- Test scenarios and test cases will be designed and executed.
- Results will be recorded and compared with expected outcomes.
- Defects will be reported and tracked until resolution.

4.4 Test Design Techniques

The following techniques will be used:

- **Equivalence Partitioning:** To test input fields with valid and invalid data
- **Boundary Value Analysis:** To test edge cases (e.g., minimum and maximum values) .

4.5 Test Tools

The following tools will be used:

- Web Browser (Google Chrome) or (Firefox).
- Different devices (Android mobile phone , iPhone mobile , desktop , tablet)
- Sauce demo Application .
- Manual Testing Techniques.
- Postman for API testing .

5. Test Environment

Testing will be performed on a Windows operating system using the Google Chrome browser with a stable internet connection. Cross-browser testing will also be conducted on Firefox to ensure compatibility across different browsers. The testing process will cover both the web and the mobile application, including Android and iOS platforms, to ensure consistent functionality, responsiveness, and user experience across all supported devices .

this environment simulates real-world usage conditions to validate system performance, compatibility, and stability across multiple platforms and devices.

6. Features to be Tested

The following features/modules of the Sauce Demo will be tested:

- User Login functionality (valid and invalid login scenarios) .
- Logout functionality .
- Product inventory display (product list page) .
- Product sorting and filtering options .
- Product details view .
- Add to Cart functionality .
- Remove from Cart functionality .
- Shopping Cart page (viewing selected items) .
- Checkout process (Step One: User Information) .
- Checkout process (Step Two: Order overview) .
- Order completion and confirmation page .

7. Test Scenarios

The following test scenarios will be covered in this project:

1. User Login

Verify that users can successfully log in using valid credentials and are prevented from logging in using invalid credentials.

2. Product Gallery & Sorting

Verify that all products are displayed correctly in the product gallery and that sorting options (e.g., price, name) work as expected.

3. Shopping Cart Management

Verify that users can add products to the cart, view cart items correctly, and remove items from the cart.

4. Checkout Process & Information

Verify that the checkout process works correctly, including entering user information, reviewing order details, and completing the purchase successfully.

5. Application Logout

Verify that users can successfully log out from the application and are redirected to the login page.

8. Defect Management

Defects will be recorded, tracked, and managed using a defect tracking system. Each defect will include details such as description, steps to reproduce, severity, and priority. Critical issues will be reported immediately for quick resolution, and all fixed defects will be retested to ensure proper resolution. The defect lifecycle includes: New, Assigned, Open, Fixed, Retested, and Closed, ensuring effective tracking and communication between teams.

9. Test Execution Plan

Test execution will begin after completing the test case design and preparing the test environment. During this phase, all test cases will be executed according to priority, starting with critical and high-priority scenarios.

The testing activities will include:

- Writing detailed test cases based on requirements and test scenarios
- Executing different types of testing such as functional, non-functional, white box, and black box testing
- Performing UI/UX testing to ensure usability and user experience quality

- Conducting automated testing for repetitive and regression test cases to improve efficiency and accuracy
- Identifying, reporting, and tracking defects during execution
- Creating bug reports (defect reports) with full details such as steps to reproduce, expected result, and actual result
- Retesting fixed defects to ensure proper resolution and no impact on existing features

Test execution will continue until all planned test cases are completed and the application meets the required quality standard.

10. Test Deliverables

The following deliverables will be produced during the testing process:

- Test Plan document
- Test Scenarios and Test Cases
- Defect Reports
- Test Execution Reports
- Final Test Summary Report